Green's Function

A Green's function is a function that inverts a differential operator, usually in the context of differential equations. For instance, the Green's function for the Laplacian operator ∇^2 is the function that satisfies $\nabla^2 G(\mathbf{x}, \mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}')$, in this case just $G(\mathbf{x}, \mathbf{x}') = -1/4\pi|\mathbf{x} - \mathbf{x}'|$. As we will see, in quantum mechanics, propagators serve as Green's functions of Schrödinger's equation.

To see how Green's functions are useful, consider some generic inhomogeneous differential equation of the form

$$\hat{L}_x \psi(x) = \rho(x), \tag{1}$$

where $\hat{L}_x$ is a linear differential operator acting on functions in the x domain, $\psi(x)$ is the function we want to solve for, and $\rho(x)$ is a known source term. This is a generic differential equation that is inhomogeneous if $\rho(x) \neq 0$ (in the $\rho(x) = 0$ homogeneous case, Green's functions are a less useful concept to talk about). For example, the Schrödinger equation can be written in this form, with $\hat{L}_x = i\hbar \frac{\partial}{\partial t} - \hat{H}$ (in its most primitive form, Schrödinger;s equation is homogeneous, but we will get to that).

The Green's function $G(x, x')$ associated with the operator $\hat{L}_x$ is defined as the solution to

$$\hat{L}_x G(x, x') = \delta(x - x'). \tag{2}$$

In reality, these usually show up inside an integral. Multiplying both sides by some arbitrary function $f(x')$ and integrating over x', we get

$$\int dx' \, \hat{L}_x G(x, x') f(x') = \int dx' \, \delta(x - x') f(x') = f(x). \tag{3}$$

So far, we have used the definition of the Green's function and seen how it inverts the operator $\hat{L}_x$. But note that the $\hat{L}_x$ operator is acting on x, while the Green's function depends on both x and x', the latter being a dummy integration variable. Because we assume $\hat{L}_x$ is some linear differential operator, we can pull it out of the integral and use $f \to \rho$ to get the much more illuminating expression

$$\hat{L}_x \left(\int dx' \, G(x, x') \rho(x') \right) = \rho(x), \tag{4}$$

which comparing with Eq. (1), shows that a solution to the inhomogeneous differential equation is simply

$$\psi(x) = \int dx' \, G(x, x') \rho(x'). \tag{5}$$

This may look like we are just shifting the problem around, since $G(x, x')$ may be just as hard to find as $\psi(x)$. However, Green's functions are often easier to find than the full solution $\psi(x)$, especially when $\rho(x)$ is a simple function (often just a delta function, for example).

In your studies of electromagnetism, you may have encountered Green's functions when solving Poisson's equation for the electric potential $V(\mathbf{x})$ due to a charge distribution $\rho(\mathbf{x})$:

$$\nabla^2 V(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon_0}. \tag{6}$$

Long story short, you probably got Green's function by looking at $r = |\mathbf{x} - \mathbf{x}'| > 0$ and solving

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}') \implies \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dG}{dr}\right) = 0 \implies G(\mathbf{x}, \mathbf{x}') = -\frac{1}{4\pi r}. \tag{7}$$

Using Eq. (5), the solution for the electric potential can be expressed as

$$V(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int d^3\mathbf{x}' \, \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \tag{8}$$

Note what this solution says: the potential at point $\mathbf{x}$ is obtained by integrating the contributions from all source points $\mathbf{x}'$, weighted by the Green's function $G(\mathbf{x}, \mathbf{x}') = -\frac{1}{4\pi|\mathbf{x}-\mathbf{x}'|}$. The latter encapsulates the influence of a single point charge located at $\mathbf{x}'$. We can say a point charge because by the comparison:

$$\nabla^2 V(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon_0} \quad \leftrightarrow \quad \nabla^2 G(\mathbf{x}, \mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}'). \tag{9}$$

we can interpret the Green's function as the potential due to a charge source of $\rho(\mathbf{x}) = q\delta(\mathbf{x} - \mathbf{x}')$.

In classical mechanics, you probably also saw this method. There you could have a force applied throughout time $m\frac{d^2x(t)}{dt^2} = F(t)$, which is also an inhomogeneous differential equation and hence the Green's function method applies,

$$x(t) = \int dt' \, G(t, t')F(t'), \quad \text{with} \quad m\frac{d^2}{dt^2}G(t, t') = \delta(t - t'). \tag{10}$$

The solutions to the equations of motion are a convolution of the real force we care about with the Green's function, which in this case captures the response of the system to an impulsive force at time t'.

The Propagator as a Green's Function

In the context of quantum mechanics, the propagator $K(x_1, t_1; x_0, t_0)$ (or something very close to how we have been using it) serves as the Green's function for the time-dependent Schrödinger equation. You should protest at this point because the Schrödinger equation is homogeneous (i.e., $(i\hbar\frac{\partial}{\partial t} - \hat{H})\psi = 0$), and the same is true for the Schrödinger equation for the propagator. However, it is not until we split the unphysical asymetry in forward and backward time propagation in our propagator that we see the real Green's function appear. This is done by splitting our propagator into a retarded and an advanced part, which will each individually satisfy an inhomogeneous equation with a delta function source term. This step is necessary to ensure causality in the solutions we obtain.[1]

[1] Non-relativistic quantum mechanics is causal, even if not locally-causal as in relativistic theories, meaning that our causality requirement cares only about the arrow of time, and not about distances.

So, let us begin with the time-dependent Schrödinger equation for the propagator:

$$\left(i\hbar\frac{\partial}{\partial t} - \hat{H}(\mathbf{x},t)\right) K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = 0. \tag{11}$$

We now define the retarded and advanced propagators, which are non-zero only for $t > t'$ and $t < t'$, respectively:

$$K_R(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = \Theta(t_1-t_0)K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0), \quad K_A(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = \Theta(t_0-t_1)K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0). \tag{12}$$

The retarded propagator K_R describes the propagation of the wavefunction forward in time, while the advanced propagator K_A describes propagation backward in time. The task now is to find what differential equation these propagators satisfy. To enforce causality, we will only consider K_R. Differentiating K_R with respect to t_1, we get

$$\frac{\partial}{\partial t_1}K_R(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = \delta(t_1-t_0)K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) + \Theta(t_1-t_0)\frac{\partial}{\partial t_1}K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0). \tag{13}$$

Using the Schrödinger equation for K, we can rewrite this as

$$\left(i\hbar\frac{\partial}{\partial t_1} - \hat{H}(\mathbf{x}_1,t_1)\right) K_R(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = i\hbar\delta(t_1-t_0)K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0). \tag{14}$$

Using $\delta(t_1-t_0)K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = \delta(t_1-t_0)\delta(\mathbf{x}_1-\mathbf{x}_0)$, we arrive at

$$\left(i\hbar\frac{\partial}{\partial t_1} - \hat{H}(\mathbf{x}_1,t_1)\right) \frac{K_R(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0)}{i\hbar} = \delta(t_1-t_0)\delta(\mathbf{x}_1-\mathbf{x}_0), \tag{15}$$

which is of the form $\hat{L}G = \delta$ we were looking for. Note that the delta function in this case is in all independent variables t_1 and $\mathbf{x}_1$, since our differential operator is defined in both the t_1 and $\mathbf{x}_1$ domains. The time and space variables with subscript 0 are, as far as the propagator and Green's function are concerned, just constants. Eventually, they will be demoted to dummy integration variables, but they show up here because our problem is phrased as a evolution from an initial state (boundary condition). To summarize, we can identify the Green's function for the time-dependent Schrödinger equation as

$$\boxed{G(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0) = \Theta(t_1-t_0)\frac{K(\mathbf{x}_1,t_1;\mathbf{x}_0,t_0)}{i\hbar}}. \tag{16}$$

A similar derivation can be done for the advanced propagator, leading to a Green's function that can help us find solutions that propagate backward in time. This step is not physical for qunatum mechanics, but it becomes crucial in quantum field theory (for antiparticles), where both retarded and advanced Green's functions play equally important roles in ensuring the theory remains locally causal.

Let us consider the physical intuition behind Eq. (16). First, we note that the wavefunction evolution is just:

$$\psi(\mathbf{x}_1, t_1) = \int d^3x_0\, K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)\psi(\mathbf{x}_0, t_0) = i\hbar \int d^3x_0\, G(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)\psi(\mathbf{x}_0, t_0). \quad (17)$$

We then recall that the wavefunction and the propagator satisfy the Schrödinger equation. The Green's function also satisfies the Schrödinger equation provided $t_1 > t_0$ (since $\delta(t_1 - t_0) = 0$ for $t_1 \neq t_0$). So, using $\hat{L}_\mathrm{S} = i\hbar\frac{\partial}{\partial t} - \hat{H}$, we have

$$\hat{L}_\mathrm{S}\psi(\mathbf{x}_1, t_1) = 0, \quad \hat{L}_\mathrm{S}K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = 0, \text{ and } \quad \hat{L}_\mathrm{S}G(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = 0. \quad (18)$$

So, just as before in E&M and classical mechanics, we can interpret the Green's function as a specific solution to our differential equation under extreme conditions: a point-like "disturbance" at position $\mathbf{x}_1$ and time t_1. In our case, we do not have a source term to interpret it as an "impulse" or "point charge", but noting that

$$G(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) \xrightarrow{(t_1 - t_0) \to 0^+} \delta(\mathbf{x}_1 - \mathbf{x}_0), \quad (19)$$

we can say that the Green's function captures the time evolution of an infinitely-pinched wavefunction at $\mathbf{x}_0$ at time t_0. Then, by convolving our initial wavefunction with this Green's function, we can time evolve any kind of wavefunction we are interested in.